

Joint Effects of PM_{2.5} Pollution and Cold Waves Exposure on Population Health Risks in Shijiazhuang, China

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Table.S1 Daily lag effect of Non-accidental Mortality under different thresholds of joint exposure
(lags 0-21 days)

Lag	P1T1	P1T2	P1T3	P2T1	P2T2	P2T3	P3T1	P3T2	P3T3	P4T1	P4T2	P4T3
0	1.02 (0.98,1.08)	1.07 (1.01,1.14)	1.07 (1.00,1.14)	1.01 (0.96,1.07)	1.05 (0.99,1.12)	1.08 (0.99,1.17)	1.01 (0.94,1.08)	1.06 (0.98,1.14)	1.06 (0.98,1.14)	1.05 (0.97,1.13)	1.06 (0.98,1.15)	1.08 (0.91,1.29)
1	1.03 (1.00,1.06)	1.06 (1.03,1.10)	1.05 (1.01,1.09)	1.03 (0.99,1.06)	1.05 (1.01,1.09)	1.05 (1.00,1.10)	1.03 (0.99,1.07)	1.06 (1.02,1.11)	1.05 (1.00,1.10)	1.06 (1.01,1.11)	1.06 (1.01,1.12)	1.04 (0.93,1.16)
2	1.03 (1.01,1.06)	1.05 (1.02,1.08)	1.04 (1.01,1.07)	1.04 (1.01,1.06)	1.05 (1.02,1.08)	1.03 (0.99,1.06)	1.04 (1.01,1.07)	1.06 (1.03,1.10)	1.04 (1.01,1.08)	1.06 (1.03,1.10)	1.06 (1.02,1.10)	1.00 (0.92,1.09)
3	1.03 (1.01,1.06)	1.04 (1.01,1.07)	1.02 (0.99,1.06)	1.04 (1.01,1.07)	1.04 (1.01,1.08)	1.01 (0.97,1.06)	1.05 (1.01,1.09)	1.06 (1.02,1.10)	1.04 (0.99,1.08)	1.06 (1.02,1.11)	1.05 (1.01,1.10)	0.98 (0.89,1.08)
4	1.03 (1.00,1.06)	1.03 (0.99,1.06)	1.02 (0.98,1.06)	1.04 (1.01,1.07)	1.03 (1.00,1.07)	1.01 (0.97,1.06)	1.05 (1.01,1.08)	1.05 (1.01,1.09)	1.03 (0.99,1.08)	1.05 (1.01,1.10)	1.04 (1.00,1.09)	0.97 (0.88,1.07)
5	1.02 (1.00,1.04)	1.01 (0.99,1.04)	1.02 (0.99,1.05)	1.02 (1.00,1.05)	1.02 (0.99,1.04)	1.01 (0.98,1.05)	1.03 (1.00,1.06)	1.03 (1.00,1.06)	1.03 (1.00,1.07)	1.04 (1.00,1.07)	1.03 (0.99,1.06)	0.97 (0.90,1.06)
6	1.01 (1.00,1.03)	1.00 (0.98,1.02)	1.02 (1.00,1.04)	1.01 (0.99,1.03)	1.00 (0.98,1.03)	1.02 (1.00,1.05)	1.02 (1.00,1.04)	1.01 (0.99,1.04)	1.03 (1.00,1.06)	1.02 (0.99,1.05)	1.01 (0.98,1.05)	0.98 (0.91,1.06)
7	1.01 (0.99,1.03)	1.00 (0.97,1.02)	1.02 (0.99,1.05)	1.00 (0.98,1.02)	0.99 (0.97,1.02)	1.03 (1.00,1.06)	1.01 (0.98,1.03)	1.00 (0.97,1.03)	1.03 (1.00,1.06)	1.01 (0.97,1.04)	1.00 (0.97,1.04)	1.00 (0.92,1.07)
8	1.01 (0.98,1.03)	0.99 (0.97,1.02)	1.02 (0.99,1.05)	1.00 (0.97,1.02)	0.99 (0.96,1.02)	1.03 (1.00,1.07)	1.00 (0.97,1.03)	0.99 (0.96,1.02)	1.03 (1.00,1.06)	1.00 (0.97,1.04)	1.00 (0.96,1.04)	1.01 (0.93,1.09)
9	1.01 (0.99,1.03)	1.00 (0.97,1.02)	1.02 (0.99,1.05)	1.00 (0.97,1.02)	0.99 (0.96,1.02)	1.03 (1.00,1.07)	1.00 (0.97,1.03)	0.99 (0.96,1.02)	1.03 (0.99,1.06)	1.00 (0.97,1.04)	1.00 (0.97,1.04)	1.02 (0.94,1.10)
10	1.01 (0.99,1.03)	1.00 (0.98,1.02)	1.02 (1.00,1.04)	1.00 (0.98,1.02)	1.00 (0.98,1.02)	1.03 (1.00,1.06)	1.00 (0.98,1.02)	1.00 (0.97,1.02)	1.02 (0.99,1.05)	1.01 (0.98,1.04)	1.01 (0.98,1.04)	1.03 (0.96,1.10)
11	1.01 (1.00,1.03)	1.01 (0.99,1.03)	1.02 (1.00,1.04)	1.01 (0.99,1.03)	1.01 (0.99,1.02)	1.03 (1.00,1.05)	1.01 (0.99,1.03)	1.00 (0.98,1.02)	1.02 (0.99,1.04)	1.02 (0.99,1.04)	1.02 (1.00,1.05)	1.04 (0.97,1.10)
12	1.02 (1.00,1.03)	1.02 (1.00,1.04)	1.02 (1.00,1.03)	1.02 (1.00,1.03)	1.02 (1.00,1.03)	1.02 (1.00,1.04)	1.01 (1.00,1.03)	1.01 (0.99,1.03)	1.01 (0.99,1.04)	1.03 (1.00,1.05)	1.03 (1.01,1.06)	1.04 (0.99,1.10)
13	1.02 (1.01,1.04)	1.03 (1.01,1.05)	1.02 (1.00,1.04)	1.03 (1.01,1.04)	1.03 (1.01,1.05)	1.02 (0.99,1.04)	1.02 (1.00,1.04)	1.02 (1.00,1.04)	1.01 (0.99,1.04)	1.04 (1.01,1.07)	1.04 (1.01,1.07)	1.05 (0.99,1.12)
14	1.03 (1.01,1.05)	1.04 (1.01,1.06)	1.02 (0.99,1.04)	1.03 (1.01,1.05)	1.03 (1.01,1.06)	1.01 (0.99,1.04)	1.03 (1.01,1.05)	1.03 (1.00,1.06)	1.01 (0.99,1.04)	1.05 (1.02,1.08)	1.05 (1.02,1.08)	1.06 (0.99,1.13)
15	1.03 (1.01,1.05)	1.04 (1.02,1.06)	1.02 (0.99,1.04)	1.03 (1.01,1.06)	1.04 (1.01,1.06)	1.01 (0.99,1.04)	1.03 (1.01,1.06)	1.04 (1.01,1.06)	1.02 (0.99,1.04)	1.05 (1.02,1.08)	1.05 (1.02,1.09)	1.06 (1.00,1.14)
16	1.03 (1.01,1.05)	1.04 (1.02,1.06)	1.02 (1.00,1.04)	1.03 (1.01,1.05)	1.04 (1.02,1.06)	1.02 (0.99,1.04)	1.04 (1.01,1.06)	1.04 (1.02,1.07)	1.02 (1.00,1.05)	1.05 (1.02,1.08)	1.05 (1.02,1.08)	1.07 (1.01,1.14)
17	1.03 (1.01,1.04)	1.04 (1.02,1.06)	1.02 (1.01,1.04)	1.03 (1.01,1.05)	1.04 (1.02,1.06)	1.02 (1.00,1.04)	1.04 (1.02,1.05)	1.04 (1.02,1.06)	1.03 (1.01,1.05)	1.05 (1.02,1.08)	1.05 (1.02,1.08)	1.08 (1.02,1.14)
18	1.03 (1.01,1.04)	1.03 (1.02,1.05)	1.03 (1.01,1.05)	1.03 (1.01,1.04)	1.03 (1.02,1.05)	1.02 (1.00,1.04)	1.03 (1.02,1.05)	1.04 (1.02,1.06)	1.04 (1.01,1.06)	1.05 (1.02,1.07)	1.04 (1.02,1.07)	1.08 (1.03,1.14)
19	1.02 (1.00,1.04)	1.03 (1.01,1.05)	1.03 (1.01,1.06)	1.02 (1.00,1.04)	1.03 (1.01,1.05)	1.03 (1.00,1.06)	1.03 (1.01,1.05)	1.04 (1.02,1.07)	1.05 (1.02,1.07)	1.04 (1.01,1.07)	1.04 (1.01,1.07)	1.09 (1.02,1.16)
20	1.02 (0.99,1.04)	1.02 (1.00,1.05)	1.04 (1.01,1.07)	1.01 (0.99,1.04)	1.02 (0.99,1.05)	1.04 (1.00,1.08)	1.03 (1.00,1.06)	1.04 (1.01,1.08)	1.06 (1.02,1.10)	1.03 (0.99,1.07)	1.03 (0.99,1.07)	1.10 (1.01,1.20)
21	1.01 (0.98,1.05)	1.02 (0.98,1.06)	1.04 (1.00,1.09)	1.01 (0.97,1.05)	1.02 (0.97,1.06)	1.04 (0.99,1.10)	1.02 (0.98,1.07)	1.04 (0.99,1.09)	1.07 (1.02,1.12)	1.03 (0.98,1.08)	1.02 (0.97,1.08)	1.10 (0.99,1.24)

Table.S2 Daily lag effect of Circulatory disease Mortality under different thresholds of joint exposure
(lags 0-21 days)

Lag	P1T1	P1T2	P1T3	P2T1	P2T2	P2T3	P3T1	P3T2	P3T3	P4T1	P4T2	P4T3
0	1.05 (0.99,1.11)	1.11 (1.04,1.19)	1.12 (1.03,1.21)	1.02 (0.95,1.09)	1.07 (0.99,1.16)	1.08 (0.99,1.17)	1.01 (0.93,1.09)	1.06 (0.97,1.16)	1.06 (0.96,1.16)	1.04 (0.95,1.14)	1.06 (0.97,1.17)	1.03 (0.93,1.15)
1	1.04 (1.01,1.08)	1.08 (1.04,1.12)	1.07 (1.03,1.12)	1.04 (1.00,1.07)	1.06 (1.02,1.11)	1.05 (1.00,1.10)	1.03 (0.98,1.08)	1.06 (1.01,1.11)	1.05 (0.99,1.11)	1.06 (1.00,1.12)	1.06 (1.00,1.13)	1.03 (0.97,1.10)
2	1.04 (1.01,1.06)	1.05 (1.02,1.09)	1.03 (1.00,1.07)	1.05 (1.02,1.08)	1.05 (1.02,1.09)	1.03 (0.99,1.06)	1.05 (1.01,1.08)	1.06 (1.02,1.10)	1.04 (1.00,1.08)	1.07 (1.02,1.12)	1.06 (1.01,1.11)	1.03 (0.99,1.08)
3	1.03 (1.00,1.07)	1.03 (0.99,1.07)	1.01 (0.97,1.06)	1.05 (1.01,1.09)	1.04 (1.00,1.08)	1.01 (0.97,1.06)	1.06 (1.01,1.10)	1.05 (1.00,1.10)	1.03 (0.99,1.09)	1.07 (1.02,1.13)	1.06 (1.00,1.11)	1.04 (0.98,1.09)
4	1.03 (0.99,1.06)	1.02 (0.98,1.05)	1.00 (0.96,1.05)	1.04 (1.01,1.08)	1.03 (0.99,1.07)	1.01 (0.97,1.06)	1.05 (1.01,1.10)	1.04 (1.00,1.09)	1.04 (0.99,1.09)	1.06 (1.01,1.12)	1.05 (1.00,1.10)	1.04 (0.99,1.10)
5	1.02 (0.99,1.05)	1.01 (0.98,1.04)	1.01 (0.98,1.04)	1.03 (1.00,1.06)	1.01 (0.98,1.05)	1.01 (0.98,1.05)	1.04 (1.01,1.08)	1.03 (0.99,1.07)	1.04 (1.00,1.08)	1.05 (1.01,1.09)	1.04 (0.99,1.08)	1.05 (1.00,1.10)
6	1.02 (1.00,1.04)	1.00 (0.98,1.03)	1.02 (0.99,1.05)	1.01 (0.99,1.04)	1.00 (0.98,1.03)	1.02 (1.00,1.05)	1.03 (1.00,1.05)	1.02 (0.99,1.05)	1.05 (1.01,1.08)	1.03 (0.99,1.07)	1.02 (0.99,1.06)	1.06 (1.01,1.10)
7	1.01 (0.99,1.04)	1.00 (0.97,1.03)	1.03 (1.00,1.06)	1.00 (0.97,1.03)	1.00 (0.97,1.03)	1.03 (1.00,1.06)	1.01 (0.98,1.04)	1.01 (0.98,1.05)	1.05 (1.01,1.08)	1.02 (0.98,1.06)	1.02 (0.98,1.06)	1.06 (1.02,1.10)
8	1.01 (0.99,1.04)	1.00 (0.97,1.04)	1.04 (1.00,1.07)	0.99 (0.97,1.02)	1.00 (0.96,1.03)	1.03 (1.00,1.07)	1.01 (0.97,1.04)	1.01 (0.97,1.04)	1.05 (1.01,1.09)	1.01 (0.97,1.05)	1.01 (0.97,1.06)	1.06 (1.01,1.11)
9	1.01 (0.99,1.04)	1.01 (0.98,1.04)	1.04 (1.01,1.07)	1.00 (0.97,1.02)	1.00 (0.97,1.03)	1.03 (1.00,1.07)	1.00 (0.97,1.04)	1.00 (0.97,1.04)	1.04 (1.00,1.08)	1.01 (0.97,1.05)	1.01 (0.97,1.06)	1.06 (1.01,1.10)
10	1.02 (0.99,1.04)	1.01 (0.99,1.04)	1.04 (1.01,1.06)	1.00 (0.98,1.03)	1.01 (0.98,1.03)	1.03 (1.00,1.06)	1.01 (0.98,1.03)	1.01 (0.98,1.04)	1.03 (1.00,1.07)	1.01 (0.98,1.05)	1.02 (0.98,1.06)	1.05 (1.01,1.09)
11	1.02 (1.00,1.04)	1.02 (1.00,1.04)	1.03 (1.01,1.05)	1.01 (0.99,1.03)	1.01 (0.99,1.04)	1.03 (1.00,1.05)	1.01 (0.99,1.03)	1.01 (0.98,1.03)	1.03 (1.00,1.05)	1.02 (0.99,1.05)	1.03 (1.00,1.06)	1.05 (1.01,1.08)
12	1.02 (1.01,1.04)	1.03 (1.01,1.05)	1.03 (1.00,1.05)	1.02 (1.00,1.04)	1.02 (1.00,1.04)	1.02 (1.00,1.04)	1.02 (0.99,1.04)	1.01 (0.99,1.04)	1.02 (0.99,1.04)	1.03 (1.00,1.06)	1.04 (1.01,1.06)	1.04 (1.01,1.08)
13	1.03 (1.01,1.05)	1.03 (1.01,1.05)	1.02 (1.00,1.04)	1.03 (1.01,1.05)	1.03 (1.01,1.05)	1.02 (0.99,1.04)	1.02 (1.00,1.05)	1.02 (0.99,1.05)	1.01 (0.98,1.04)	1.04 (1.01,1.07)	1.04 (1.01,1.08)	1.03 (1.00,1.07)
14	1.03 (1.00,1.05)	1.04 (1.01,1.06)	1.02 (0.99,1.04)	1.04 (1.01,1.06)	1.04 (1.01,1.07)	1.01 (0.99,1.04)	1.03 (1.00,1.06)	1.02 (0.99,1.06)	1.01 (0.98,1.04)	1.04 (1.01,1.08)	1.05 (1.01,1.09)	1.03 (0.99,1.07)
15	1.03 (1.00,1.05)	1.04 (1.01,1.07)	1.01 (0.99,1.04)	1.04 (1.01,1.07)	1.04 (1.01,1.07)	1.01 (0.99,1.04)	1.03 (1.00,1.06)	1.03 (1.00,1.06)	1.01 (0.98,1.04)	1.05 (1.01,1.09)	1.05 (1.01,1.09)	1.03 (0.99,1.07)
16	1.03 (1.01,1.05)	1.04 (1.01,1.06)	1.02 (0.99,1.04)	1.04 (1.01,1.06)	1.04 (1.01,1.07)	1.02 (0.99,1.04)	1.03 (1.01,1.06)	1.03 (1.00,1.06)	1.01 (0.99,1.05)	1.05 (1.01,1.08)	1.05 (1.01,1.08)	1.03 (0.99,1.07)
17	1.02 (1.01,1.04)	1.04 (1.02,1.06)	1.02 (1.00,1.04)	1.03 (1.01,1.05)	1.04 (1.02,1.06)	1.02 (1.00,1.04)	1.03 (1.01,1.06)	1.04 (1.01,1.06)	1.02 (1.00,1.05)	1.04 (1.01,1.08)	1.04 (1.01,1.07)	1.03 (0.99,1.06)
18	1.02 (1.01,1.04)	1.03 (1.01,1.05)	1.03 (1.01,1.05)	1.03 (1.01,1.04)	1.03 (1.01,1.05)	1.02 (1.00,1.04)	1.03 (1.01,1.05)	1.04 (1.02,1.06)	1.03 (1.01,1.06)	1.04 (1.01,1.07)	1.04 (1.01,1.07)	1.03 (0.99,1.07)
19	1.02 (1.00,1.04)	1.03 (1.01,1.05)	1.03 (1.01,1.06)	1.02 (0.99,1.04)	1.03 (1.00,1.05)	1.03 (1.00,1.06)	1.03 (1.00,1.06)	1.04 (1.02,1.07)	1.05 (1.01,1.08)	1.03 (1.00,1.07)	1.03 (0.99,1.07)	1.03 (0.99,1.08)
20	1.01 (0.98,1.05)	1.02 (0.99,1.06)	1.04 (1.00,1.08)	1.00 (0.97,1.04)	1.02 (0.99,1.06)	1.04 (1.00,1.08)	1.03 (0.99,1.07)	1.05 (1.01,1.09)	1.06 (1.02,1.11)	1.03(0.98, 1.07)	1.02 (0.97,1.07)	1.04 (0.98,1.09)
21	1.01 (0.97,1.05)	1.02 (0.97,1.07)	1.05 (0.99,1.11)	0.99 (0.95,1.04)	1.01 (0.96,1.06)	1.04 (0.99,1.10)	1.03 (0.97,1.08)	1.05 (0.99,1.11)	1.08 (1.02,1.14)	1.02 (0.96,1.08)	1.01 (0.95,1.08)	1.04 (0.97,1.12)

Table.S3 Daily lag effect of Respiratory disease Mortality under different thresholds of joint exposure
(lags 0-21 days)

Lag	P1T1	P1T2	P1T3	P2T1	P2T2	P2T3	P3T1	P3T2	P3T3	P4T1	P4T2	P4T3
0	0.98 (0.89,1.09)	1.06 (0.94,1.19)	1.04 (0.91,1.20)	1.00 (0.89,1.12)	1.05 (0.93,1.20)	1.04 (0.90,1.20)	1.04 (0.91,1.19)	1.09 (0.94,1.27)	1.06 (0.90,1.25)	1.12 (0.96,1.31)	1.14 (0.97,1.34)	1.08 (0.91,1.29)
1	0.99 (0.94,1.05)	1.05 (0.98,1.12)	1.04 (0.96,1.12)	0.99 (0.93,1.05)	1.04 (0.97,1.12)	1.03 (0.95,1.11)	1.02 (0.95,1.11)	1.08 (0.99,1.18)	1.05 (0.96,1.15)	1.06 (0.97,1.17)	1.08 (0.98,1.19)	1.04 (0.93,1.16)
2	1.00 (0.96,1.04)	1.03 (0.97,1.09)	1.03 (0.97,1.09)	0.98 (0.93,1.03)	1.03 (0.97,1.09)	1.02 (0.96,1.08)	1.01 (0.95,1.07)	1.06 (0.99,1.14)	1.04 (0.97,1.12)	1.01 (0.94,1.09)	1.03 (0.95,1.12)	1.00 (0.92,1.09)
3	1.00 (0.95,1.06)	1.01 (0.95,1.08)	1.02 (0.95,1.10)	0.98 (0.92,1.04)	1.01 (0.94,1.09)	1.01 (0.93,1.09)	0.99 (0.92,1.07)	1.04 (0.96,1.13)	1.03 (0.94,1.12)	0.98 (0.89,1.07)	0.99 (0.91,1.09)	0.98 (0.89,1.08)
4	1.00 (0.95,1.06)	1.00 (0.93,1.06)	1.01 (0.94,1.08)	0.98 (0.92,1.04)	1.00 (0.93,1.07)	1.00 (0.92,1.08)	0.98 (0.91,1.06)	1.02 (0.94,1.10)	1.01 (0.93,1.11)	0.96 (0.88,1.05)	0.97 (0.89,1.06)	0.97 (0.88,1.07)
5	1.00 (0.96,1.04)	0.98 (0.93,1.03)	0.99 (0.94,1.05)	0.99 (0.94,1.03)	0.98 (0.93,1.04)	0.99 (0.93,1.05)	0.98 (0.93,1.04)	0.99 (0.93,1.06)	1.00 (0.93,1.07)	0.97 (0.90,1.04)	0.96 (0.89,1.04)	0.97 (0.90,1.06)
6	1.00 (0.97,1.03)	0.97 (0.93,1.01)	0.98 (0.94,1.03)	0.99 (0.95,1.03)	0.97 (0.93,1.02)	0.98 (0.94,1.03)	0.98 (0.94,1.03)	0.97 (0.92,1.02)	0.99 (0.93,1.05)	0.98 (0.92,1.04)	0.97 (0.90,1.03)	0.98 (0.91,1.06)
7	1.00 (0.96,1.04)	0.96 (0.92,1.01)	0.98 (0.93,1.03)	1.00 (0.96,1.05)	0.97 (0.92,1.02)	0.98 (0.93,1.04)	0.99 (0.94,1.04)	0.96 (0.90,1.01)	0.98 (0.92,1.04)	1.00 (0.93,1.06)	0.98 (0.91,1.05)	1.00 (0.92,1.07)
8	1.00 (0.96,1.04)	0.97 (0.92,1.02)	0.97 (0.92,1.03)	1.01 (0.96,1.06)	0.97 (0.91,1.03)	0.98 (0.92,1.04)	1.00 (0.94,1.05)	0.95 (0.90,1.02)	0.98 (0.91,1.05)	1.01 (0.94,1.09)	0.99 (0.92,1.06)	1.01 (0.93,1.09)
9	1.00 (0.96,1.05)	0.97 (0.93,1.02)	0.98 (0.92,1.03)	1.01 (0.97,1.06)	0.98 (0.92,1.03)	0.98 (0.93,1.04)	1.00 (0.95,1.06)	0.96 (0.91,1.02)	0.98 (0.92,1.05)	1.03 (0.96,1.10)	1.00 (0.93,1.08)	1.02 (0.94,1.10)
10	1.01 (0.97,1.04)	0.99 (0.95,1.03)	0.98 (0.94,1.03)	1.02 (0.98,1.06)	0.99 (0.94,1.03)	0.99 (0.94,1.04)	1.01 (0.97,1.06)	0.98 (0.93,1.03)	0.99 (0.93,1.05)	1.04 (0.98,1.10)	1.02 (0.96,1.08)	1.03 (0.96,1.10)
11	1.01 (0.98,1.04)	1.00 (0.97,1.04)	0.99 (0.95,1.03)	1.02 (0.99,1.06)	1.00 (0.97,1.04)	0.99 (0.96,1.03)	1.02 (0.99,1.06)	1.00 (0.96,1.05)	1.00 (0.95,1.05)	1.05 (1.00,1.10)	1.04 (0.99,1.09)	1.04 (0.97,1.10)
12	1.02 (0.99,1.05)	1.02 (0.99,1.05)	1.00 (0.97,1.04)	1.02 (0.99,1.06)	1.02 (0.99,1.05)	1.00 (0.97,1.04)	1.03 (1.00,1.07)	1.03 (0.99,1.07)	1.01 (0.97,1.06)	1.06 (1.01,1.11)	1.05 (1.00,1.10)	1.04 (0.99,1.10)
13	1.03 (0.99,1.06)	1.03 (1.00,1.07)	1.01 (0.97,1.05)	1.03 (0.99,1.06)	1.03 (0.99,1.07)	1.01 (0.97,1.05)	1.04 (1.00,1.08)	1.05 (1.00,1.09)	1.03 (0.98,1.08)	1.06 (1.01,1.12)	1.07 (1.01,1.12)	1.05 (0.99,1.12)
14	1.03 (0.99,1.07)	1.05 (1.00,1.09)	1.02 (0.98,1.07)	1.03 (0.99,1.07)	1.04 (1.00,1.09)	1.02 (0.97,1.07)	1.05 (1.00,1.10)	1.07 (1.01,1.12)	1.04 (0.98,1.09)	1.07 (1.01,1.13)	1.08 (1.01,1.14)	1.06 (0.99,1.13)
15	1.03 (1.00,1.07)	1.05 (1.01,1.10)	1.03 (0.98,1.08)	1.03 (0.99,1.08)	1.05 (1.00,1.10)	1.03 (0.98,1.08)	1.05 (1.00,1.10)	1.08 (1.02,1.13)	1.05 (0.99,1.11)	1.07 (1.01,1.14)	1.08 (1.02,1.15)	1.06 (1.00,1.14)
16	1.04 (1.00,1.07)	1.06 (1.02,1.10)	1.04 (0.99,1.08)	1.03 (0.99,1.07)	1.06 (1.01,1.10)	1.03 (0.99,1.08)	1.05 (1.01,1.10)	1.08 (1.03,1.13)	1.05 (1.00,1.11)	1.08 (1.02,1.14)	1.08 (1.02,1.15)	1.07 (1.01,1.14)
17	1.04 (1.01,1.07)	1.05 (1.02,1.09)	1.04 (1.01,1.08)	1.04 (1.00,1.07)	1.05 (1.02,1.09)	1.04 (1.00,1.08)	1.05 (1.01,1.09)	1.07 (1.03,1.12)	1.06 (1.02,1.11)	1.08 (1.03,1.13)	1.08 (1.03,1.14)	1.08 (1.02,1.14)
18	1.04 (1.01,1.07)	1.05 (1.02,1.08)	1.05 (1.01,1.08)	1.04 (1.01,1.07)	1.05 (1.02,1.08)	1.05 (1.01,1.08)	1.05 (1.01,1.09)	1.07 (1.03,1.11)	1.06 (1.02,1.11)	1.08 (1.03,1.13)	1.08 (1.03,1.13)	1.08 (1.03,1.14)
19	1.04 (1.00,1.07)	1.04 (1.01,1.08)	1.05 (1.01,1.10)	1.04 (1.00,1.08)	1.05 (1.01,1.09)	1.05 (1.01,1.10)	1.05 (1.00,1.09)	1.05 (1.01,1.10)	1.07 (1.01,1.12)	1.08 (1.02,1.14)	1.07 (1.01,1.13)	1.09 (1.02,1.16)
20	1.04 (0.99,1.09)	1.04 (0.98,1.09)	1.05 (0.99,1.12)	1.05 (0.99,1.11)	1.04 (0.98,1.10)	1.06 (0.99,1.13)	1.04 (0.98,1.11)	1.04 (0.98,1.11)	1.07 (1.00,1.15)	1.07 (1.00,1.16)	1.06 (0.99,1.15)	1.10 (1.01,1.20)
21	1.04 (0.97,1.11)	1.03 (0.95,1.11)	1.06 (0.97,1.15)	1.05 (0.97,1.13)	1.03 (0.95,1.12)	1.07 (0.97,1.16)	1.04 (0.95,1.13)	1.03 (0.94,1.12)	1.07 (0.97,1.18)	1.07 (0.97,1.19)	1.06 (0.95,1.17)	1.10 (0.99,1.24)

Table.S4 Cumulative Relative Risks and Attributable Burden Metrics of PM_{2.5}-only, Cold-only, and Joint Exposures for Non-accidental Mortality

Exposure Combination Definition	RR1 (95% CI)	RR2 (95% CI)	RR3 (95% CI)	AF (95% CI)	AN (95% CI)
P1T1	1.09 (0.92,1.28)	1.42 (1.27,1.58)***	1.73 (1.43,2.10)***	42.2% (30.0%-52.3%)	1696 (1204-2102)
P1T2	1.05 (0.89,1.24)	1.30 (1.14,1.48)***	1.87 (1.52,2.31)***	46.5% (34.1%-56.6%)	1402 (1027-1707)
P1T3	1.16 (0.99,1.37)	1.71 (1.46,1.99)***	2.01 (1.59,2.55)***	50.2% (37.0%-60.7%)	1049 (772-1268)
P2T1	1.01 (0.83,1.21)	1.42 (1.28,1.58)***	1.75 (1.43,2.15)***	42.9% (29.9%-53.4%)	1409 (984-1755)
P2T2	0.91 (0.76,1.10)	1.30 (1.15,1.48)***	1.72 (1.39,2.13)***	41.9% (28.0%-53.0%)	1113 (746-1411)
P2T3	1.04 (0.86,1.26)	1.70 (1.46,1.98)***	1.88 (1.48,2.39)***	46.8% (32.4%-58.1%)	885 (612-1099)
P3T1	1.10 (0.89,1.36)	1.41 (1.28,1.56)***	1.94 (1.53,2.46)***	48.5% (34.6%-59.3%)	1089 (778-1334)
P3T2	0.98 (0.80,1.21)	1.30 (1.15,1.47)***	1.97 (1.54,2.53)***	49.2% (34.9%-60.4%)	954 (677-1170)
P3T3	1.11 (0.90,1.36)	1.68 (1.45,1.94)***	1.98 (1.46,2.70)***	49.5% (31.3%-62.9%)	664 (420-844)
P4T1	1.31 (0.91,1.90)	1.42 (1.28,1.57)***	2.55 (1.82,3.57)***	60.8% (45.1%-72.0%)	883 (655-1045)
P4T2	1.21 (0.85,1.72)	1.33 (1.18,1.50)***	2.15 (1.52,3.05)***	53.5% (34.1%-67.2%)	712 (454-895)
P4T3	1.56 (1.10,2.20)*	1.73 (1.50,2.01)***	1.75 (1.10,2.80)*	42.9% (8.8%-64.2%)	368 (76-551)

Note: RR¹: cumulative relative risk for PM_{2.5}-only events; RR²: cumulative relative risk for cold waves-only events; RR³: cumulative relative risk for joint exposure events to PM_{2.5} and cold waves. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

Table.S5 Cumulative Relative Risks and Attributable Burden Metrics of PM_{2.5}-only, Cold-only, and Joint Exposures for Circulatory disease Mortality

Exposure Combination Definition	RR ¹ (95% CI)	RR ² (95% CI)	RR ³ (95% CI)	AF (95% CI)	AN (95% CI)
P1T1	1.07 (0.88,1.31)	1.33 (1.17,1.52)***	1.81 (1.44,2.26)***	44.8% (30.8%-55.9%)	994 (684-1242)

P1T2	1.03 (0.85,1.25)	1.27 (1.09,1.47)**	1.99 (1.56,2.55)***	49.7% (35.8%-60.7%)	846 (608-1032)
P1T3	1.15 (0.95,1.39)	1.69 (1.41,2.03)***	2.24 (1.70,2.97)***	55.4% (41.0%-66.2%)	653 (484-781)
P2T1	1.02 (0.81,1.28)	1.35 (1.19,1.53)***	1.86 (1.46,2.37)***	46.2% (31.5%-57.8%)	836 (570-1045)
P2T2	0.93 (0.74,1.16)	1.29 (1.11,1.50)**	1.84 (1.42,2.37)***	45.7% (29.8%-57.9%)	677 (442-859)
P2T3	1.06 (0.85,1.33)	1.71 (1.43,2.05)***	2.10 (1.58,2.79)***	52.4% (36.7%-64.2%)	552 (387-676)
P3T1	1.08 (0.84,1.39)	1.35 (1.20,1.53)***	2.03 (1.52,2.69)***	50.7% (34.5%-63.0%)	636 (432-790)
P3T2	0.96 (0.75,1.24)	1.29 (1.12,1.49)***	2.03 (1.51,2.73)***	50.7% (33.8%-63.4%)	552 (367-689)
P3T3	1.10 (0.86,1.40)	1.70 (1.43,2.02)***	2.15 (1.49,3.10)***	53.5% (32.9%-67.8%)	398 (245-504)
P4T1	1.16 (0.74,1.81)	1.35 (1.20,1.53)***	2.62 (1.76,3.92)***	61.8% (43.0%-74.4%)	515 (359-620)
P4T2	1.06 (0.69,1.62)	1.31 (1.14,1.52)***	2.12 (1.40,3.21)***	52.8% (28.6%-68.8%)	403 (218-525)
P4T3	1.45 (0.95,2.20)	1.76 (1.48,2.10)***	1.83 (1.05,3.20)*	45.4% (4.6%-68.7%)	220 (22-333)

Note: RR¹: cumulative relative risk for PM_{2.5}-only events; RR²: cumulative relative risk for cold waves-only events; RR³ : cumulative relative risk for joint exposure events to PM_{2.5} and cold waves. **P* < 0.05, ***P* < 0.01, ****P* < 0.001.

Table.S6 Cumulative Relative Risks and Attributable Burden Metrics of PM_{2.5}-only, Cold-only, and Joint Exposures for Respiratory disease Mortality

Exposure Combination Definition	RR ¹ (95% CI)	RR ² (95% CI)	RR ³ (95% CI)	AF (95% CI)	AN (95% CI)
P1T1	1.09 (0.78,1.53)	1.82 (1.46,2.27)***	1.36 (0.93,1.98)	26.5% (-7.3%-49.6%)	152 (-42-285)
P1T2	1.07 (0.76,1.49)	1.36 (1.05,1.75)*	1.53 (1.00,2.35)*	34.6% (-0.2%-57.4%)	150 (-1-248)
P1T3	1.17 (0.84,1.64)	1.62 (1.19,2.20)**	1.48 (0.91,2.41)	32.4% (-10.0%-58.5%)	100 (-31-181)
P2T1	1.09 (0.74,1.61)	1.74 (1.42,2.13)***	1.48 (0.98,2.25)	32.4% (-2.4%-55.4%)	150 (-11-257)
P2T2	1.00 (0.68,1.47)	1.35 (1.05,1.73)*	1.49 (0.96,2.31)	32.9% (-4.1%-56.7%)	124 (-15-214)
P2T3	1.16 (0.78,1.71)	1.60 (1.18,2.17)**	1.42 (0.86,2.33)	29.6% (-15.9%-57.2%)	83 (-45-161)

P3T1	1.31 (0.85,2.03)	1.67 (1.37,2.04)***	1.76 (1.08,2.86)*	43.2% (7.5%-65.1%)	136 (24-204)
P3T2	1.25 (0.82,1.92)	1.31 (1.02,1.66)*	1.91 (1.15,3.18)*	47.6% (12.9%-68.5%)	126 (34-182)
P3T3	1.43 (0.94,2.19)	1.53 (1.13,2.06)**	1.57 (0.83,2.98)	36.3% (-20.7%-66.4%)	72 (-41-131)
P4T1	1.94 (0.94,4.04)	1.69 (1.39,2.07)***	2.34 (1.18,4.65)*	57.3% (15.2%-78.5%)	110 (29-151)
P4T2	1.95 (0.97,3.90)	1.33 (1.04,1.69)*	2.33 (1.15,4.73)*	57.1% (13.0%-78.8%)	99 (23-137)
P4T3	2.63 (1.31,5.28)**	1.56 (1.16,2.10)**	1.48 (0.56,3.87)	32.4% (-77.6%-74.3%)	38 (-92-88)

Note: RR¹: cumulative relative risk for PM_{2.5}-only events; RR²: cumulative relative risk for cold waves-only events; RR³: cumulative relative risk for joint exposure events to PM_{2.5} and cold waves. **P* < 0.05, ***P* < 0.01, ****P* < 0.001.

Table.S7 Interaction measures (RERI, AP, S) with 95% confidence intervals across exposure thresholds by Non-accidental Mortality

Exposure Combination Definition	RERI (95% CI)	AP (95% CI)	S (95% CI)	P
P1T1	0.23 (-0.18,0.63)	0.13 (-0.08,0.35)	1.15 (0.87,1.44)	0.243
P1T2	0.52 (0.07,0.98)	0.28 (0.08,0.48)	1.39 (1.01,1.77)	0.003
P1T3	0.14 (-0.43,0.72)	0.07 (-0.20,0.34)	1.08 (0.76,1.39)	0.898
P2T1	0.33 (-0.11,0.76)	0.19 (-0.03,0.40)	1.23 (0.90,1.55)	0.065
P2T2	0.51 (0.07,0.95)	0.30 (0.09,0.50)	1.42 (1.01,1.83)	0.001
P2T3	0.13 (-0.42,0.69)	0.07 (-0.21,0.35)	1.08 (0.75,1.40)	0.648
P3T1	0.43 (-0.11,0.96)	0.22 (-0.01,0.45)	1.28 (0.90,1.66)	0.064
P3T2	0.69 (0.13,1.24)	0.35 (0.14,0.56)	1.54 (1.05,2.03)	<0.001
P3T3	0.20 (-0.50,0.89)	0.10 (-0.23,0.42)	1.11 (0.71,1.51)	0.615
P4T1	0.82 (-0.18,1.81)	0.32 (0.02,0.62)	1.47 (0.82,2.12)	0.107
P4T2	0.62 (-0.26,1.50)	0.29 (-0.04,0.61)	1.40 (0.76,2.04)	0.118
P4T3	-0.54 (-1.56,0.47)	-0.31 (-1.01,0.39)	0.76 (0.35,1.17)	0.023

Table.S8 Interaction measures (RERI, AP, S) with 95% confidence intervals across exposure thresholds
by Circulatory disease Mortality

Exposure Combination Definition	RERI (95% CI)	AP (95% CI)	S (95% CI)	P
P1T1	0.40 (-0.09,0.89)	0.22 (-0.01,0.45)	1.29 (0.90,1.67)	0.05
P1T2	0.69 (0.13,1.25)	0.35 (0.13,0.56)	1.53 (1.03,2.03)	<0.001
P1T3	0.40 (-0.33,1.14)	0.18 (-0.10,0.46)	1.22 (0.80,1.64)	0.288
P2T1	0.49 (-0.05,1.03)	0.26 (0.03,0.50)	1.36 (0.92,1.79)	0.023
P2T2	0.62 (0.07,1.17)	0.34 (0.11,0.57)	1.51 (0.99,2.03)	0.002
P2T3	0.33 (-0.39,1.04)	0.15 (-0.15,0.46)	1.18 (0.76,1.61)	0.332
P3T1	0.59 (-0.07,1.25)	0.29 (0.04,0.55)	1.41 (0.90,1.92)	0.023
P3T2	0.77 (0.10,1.44)	0.38 (0.14,0.62)	1.61 (1.00,2.23)	<0.001
P3T3	0.36 (-0.53,1.24)	0.17 (-0.19,0.52)	1.20 (0.69,1.71)	0.341
P4T1	1.11 (-0.07,2.30)	0.43 (0.12,0.73)	1.74 (0.80,2.68)	0.028
P4T2	0.74 (-0.26,1.75)	0.35 (-0.00,0.70)	1.54 (0.70,2.38)	0.068
P4T3	-0.38 (-1.60,0.85)	-0.21 (-0.97,0.56)	0.83 (0.30,1.36)	0.153

Table.S9 Interaction measures (RERI, AP, S) with 95% confidence intervals across exposure thresholds
by Respiratory disease Mortality

Exposure Combination Definition	RERI (95% CI)	AP (95% CI)	S (95% CI)	P
P1T1	-0.56 (-1.30,0.19)	-0.41 (-1.08,0.25)	0.71 (0.37,1.04)	0.062
P1T2	0.11 (-0.71,0.93)	0.07 (-0.44,0.58)	1.08 (0.48,1.67)	0.789
P1T3	-0.32 (-1.28,0.65)	-0.21 (-0.95,0.52)	0.82 (0.33,1.32)	0.278
P2T1	-0.35 (-1.18,0.48)	-0.23 (-0.87,0.40)	0.81 (0.39,1.23)	0.273
P2T2	0.14 (-0.69,0.97)	0.10 (-0.43,0.62)	1.11 (0.46,1.75)	0.665
P2T3	-0.34 (-1.31,0.63)	-0.24 (-1.02,0.53)	0.81 (0.30,1.31)	0.288
P3T1	-0.22 (-1.30,0.86)	-0.13 (-0.79,0.54)	0.89 (0.36,1.41)	0.368

P3T2	0.36 (-0.80,1.51)	0.19 (-0.34,0.71)	1.23 (0.44,2.02)	0.53
P3T3	-0.39 (-1.65,0.87)	-0.25 (-1.18,0.68)	0.80 (0.20,1.40)	0.209
P4T1	-0.30 (-2.47,1.87)	-0.13 (-1.12,0.87)	0.89 (0.11,1.67)	0.378
P4T2	0.06 (-2.10,2.22)	0.02 (-0.89,0.94)	1.02 (0.07,1.98)	0.782
P4T3	-1.71 (-4.08,0.65)	-1.16 (-3.61,1.29)	0.46 (-0.06,0.99)	0.008

Table.S10 The results of Q-AIC of joint events definitions

Exposure Combination Definition	Non-accidental Mortality	Circulatory disease Mortality	Respiratory disease Mortality
P1T1	3486.40	3810.65	3698.35
P2T1	3490.08	3815.80	3695.76
P3T1	3498.73	3812.10	3681.32
P4T1	3499.45	3815.55	3688.26
P1T2	3434.25	3786.88	3630.95
P2T2	3429.00	3771.28	3634.06
P3T2	3432.71	3764.76	3632.75
P4T2	3417.48	3754.22	3632.50
P1T3	3477.53	3828.52	3639.03
P2T3	3476.06	3820.75	3644.00
P3T3	3482.85	3820.36	3631.66
P4T3	3481.71	3818.83	3634.01

Table.S11 Cumulative Relative Risks, Attributable Burden Metrics and Interaction Indicators of
PM_{2.5}-only, Cold-only, and Joint Exposures for Age,0-64yrs

Exposure Combination Definition	RR ¹ (95%CI)	RR ² (95%CI)	RR ³ (95%CI)	RERI (95%CI)	AP (95%CI)	S (95%CI)	AF (95%CI)	AN (95%CI)
P1T1	1.30 (1.05,1.60)*	1.50 (1.29,1.76)***	1.23 (0.97,1.56)	-0.57 (-1.03,-0.10)	-0.46 (-0.91,-0.00)	0.69 (0.47,0.90)	18.7% (-3.1%-35.9%)	163 (-27-313)
P1T2	1.28 (1.04,1.58)*	1.47 (1.25,1.73)***	1.17 (0.90,1.53)	-0.57 (-1.05,-0.10)	-0.49 (-0.99,0.01)	0.67 (0.45,0.90)	14.5% (-11.4%-34.4%)	99 (-78-236)
P1T3	1.30 (1.05,1.61)*	1.65 (1.37,2.00)***	1.20 (0.88,1.63)	-0.76 (-1.31,-0.20)	-0.63 (-1.24,-0.02)	0.61 (0.38,0.84)	16.7% (-13.4%-38.8%)	78 (-63-183)
P2T1	1.35 (1.08,1.70)**	1.47 (1.26,1.71)***	1.26 (0.99,1.61)	-0.56 (-1.05,-0.07)	-0.44 (-0.91,0.02)	0.69 (0.47,0.92)	20.6% (-1.2%-37.8%)	149 (-9-273)
P2T2	1.33 (1.07,1.65)*	1.45 (1.24,1.70)***	1.15 (0.88,1.51)	-0.63 (-1.11,-0.15)	-0.55 (-1.07,-0.02)	0.65 (0.43,0.87)	13.0% (-13.9%-33.6%)	79 (-84-204)
P2T3	1.37 (1.10,1.70)**	1.63 (1.36,1.95)***	1.16 (0.85,1.59)	-0.83 (-1.39,-0.28)	-0.71 (-1.36,-0.07)	0.58 (0.36,0.80)	13.8% (-17.9%-37.0%)	59 (-77-158)
P3T1	1.40 (1.10,1.78)**	1.46 (1.26,1.68)***	1.10 (0.82,1.47)	-0.76 (-1.26,-0.25)	-0.69 (-1.30,-0.08)	0.59 (0.38,0.81)	9.1% (-21.7%-32.1%)	46 (-110-162)
P3T2	1.33 (1.06,1.67)*	1.43 (1.24,1.66)***	1.02 (0.74,1.40)	-0.74 (-1.23,-0.25)	-0.73 (-1.39,-0.07)	0.58 (0.36,0.80)	2.0% (-34.8%-28.7%)	9 (-155-128)
P3T3	1.30 (1.04,1.62)*	1.64 (1.37,1.95)***	0.89 (0.59,1.35)	-1.04 (-1.59,-0.49)	-1.17 (-2.17,-0.16)	0.46 (0.25,0.68)	-12.4% (-70.0%-25.7%)	-38 (-213-78)
P4T1	1.46 (0.98,2.18)	1.36 (1.18,1.57)***	1.08 (0.70,1.66)	-0.74 (-1.51,0.03)	-0.69 (-1.61,0.24)	0.59 (0.27,0.92)	7.4% (-42.6%-39.9%)	25 (-145-136)
P4T2	1.39 (0.96,2.00)	1.36 (1.17,1.57)***	0.90 (0.57,1.41)	-0.84 (-1.53,-0.16)	-0.94 (-2.02,0.13)	0.51 (0.23,0.80)	-11.1% (-74.8%-29.4%)	-35 (-238-93)
P4T3	1.41 (0.97,2.03)	1.52 (1.28,1.82)***	0.75 (0.40,1.40)	-1.18 (-1.94,-0.43)	-1.59 (-3.40,0.23)	0.39 (0.12,0.66)	-33.3% (-149.4%-28.7%)	-68 (-303-58)

Note: RR¹: cumulative relative risk for PM_{2.5}-only events; RR²: cumulative relative risk for cold waves-only events; RR³ : cumulative relative risk for joint exposure events to PM_{2.5} and cold waves. **P* < 0.05, ***P* < 0.01, ****P* < 0.001.

Table.S12 Cumulative Relative Risks, Attributable Burden Metrics and Interaction Indicators of
PM_{2.5}-only, Cold-only, and Joint Exposures for Age,65-74yrs

Exposure Combination Definition	RR ¹ (95%CI)	RR ² (95%CI)	RR ³ (95%CI)	RERI (95%CI)	AP (95%CI)	S (95%CI)	AF (95%CI)	AN (95%CI)
P1T1	0.93 (0.71,1.21)	1.42 (1.18,1.70)***	1.42 (1.04,1.93)*	0.07 (-0.50,0.63)	0.05 (-0.34,0.43)	1.05 (0.62,1.48)	29.6% (4.1%-48.3%)	240 (33-392)
P1T2	0.89 (0.69,1.16)	1.34 (1.09,1.65)**	1.60 (1.15,2.23)**	0.36 (-0.28,1.01)	0.23 (-0.11,0.57)	1.30 (0.72,1.87)	37.5% (13.0%-55.1%)	239 (83-352)
P1T3	1.03 (0.79,1.33)	1.79 (1.40,2.29)***	1.60 (1.09,2.33)*	-0.21 (-1.01,0.58)	-0.13 (-0.67,0.40)	0.88 (0.46,1.30)	37.5% (8.6%-57.3%)	169 (39-258)
P2T1	0.74 (0.54,1.01)	1.36 (1.14,1.61)***	1.44 (1.04,2.00)*	0.34 (-0.23,0.92)	0.24 (-0.10,0.58)	1.31 (0.73,1.90)	30.6% (3.7%-49.9%)	207 (25-337)
P2T2	0.70 (0.51,0.95)*	1.32 (1.08,1.62)**	1.43 (1.02,2.02)*	0.41 (-0.18,1.01)	0.29 (-0.05,0.63)	1.41 (0.73,2.08)	30.1% (1.6%-50.3%)	168 (9-281)
P2T3	0.87 (0.64,1.18)	1.76 (1.38,2.25)***	1.46 (0.99,2.14)	-0.18 (-0.93,0.58)	-0.12 (-0.68,0.43)	0.89 (0.45,1.33)	31.5% (-0.7%-53.4%)	129 (-3-218)
P3T1	0.71 (0.50,1.00)	1.38 (1.16,1.62)***	1.42 (0.97,2.08)	0.34 (-0.30,0.97)	0.24 (-0.14,0.61)	1.31 (0.67,1.95)	29.6% (-3.1%-51.9%)	134 (-14-235)
P3T2	0.66 (0.47,0.92)*	1.37 (1.13,1.66)**	1.47 (0.99,2.17)	0.44 (-0.23,1.11)	0.30 (-0.06,0.66)	1.43 (0.69,2.18)	32.0% (-0.7%-54.1%)	128 (-3-216)
P3T3	0.81 (0.58,1.13)	1.81 (1.43,2.30)***	1.28 (0.78,2.11)	-0.34 (-1.16,0.47)	-0.27 (-1.01,0.47)	0.79 (0.33,1.25)	21.9% (-28.5%-52.5%)	62 (-81-150)
P4T1	0.77 (0.42,1.40)	1.40 (1.19,1.66)***	1.66 (0.96,2.86)	0.49 (-0.55,1.53)	0.30 (-0.20,0.79)	1.42 (0.42,2.42)	39.8% (-4.0%-65.1%)	120 (-12-196)
P4T2	0.75 (0.42,1.32)	1.41 (1.16,1.71)***	1.41 (0.81,2.46)	0.25 (-0.68,1.18)	0.18 (-0.41,0.76)	1.22 (0.35,2.08)	29.1% (-23.6%-59.3%)	82 (-67-167)
P4T3	1.13 (0.65,1.99)	1.88 (1.49,2.38)***	0.92 (0.43,1.97)	-1.09 (-2.14,-0.05)	-1.18 (-3.05,0.68)	0.46 (0.07,0.85)	-8.7% (-132.7%-49.2%)	-17 (-260-96)

Note: RR¹: cumulative relative risk for PM_{2.5}-only events; RR²: cumulative relative risk for cold waves-only events; RR³ : cumulative relative risk for joint exposure events to PM_{2.5} and cold waves. **P* < 0.05, ***P* < 0.01, ****P* < 0.001.

Table.S13 Cumulative Relative Risks, Attributable Burden Metrics and Interaction Indicators of
PM_{2.5}-only, Cold-only, and Joint Exposures for Age,75+yrs

Exposure Combination Definition	RR ¹ (95%CI)	RR ² (95%CI)	RR ³ (95%CI)	RERI (95%CI)	AP (95%CI)	S (95%CI)	AF (95%CI)	AN (95%CI)
P1T1	1.14 (0.95,1.36)	1.42 (1.26,1.61)***	1.82 (1.47,2.25)***	0.26 (-0.21,0.73)	0.14 (-0.09,0.38)	1.17 (0.85,1.49)	45.1% (32.0%-55.6%)	1054 (749-1301)
P1T2	1.11 (0.93,1.33)	1.25 (1.09,1.44)**	1.93 (1.53,2.44)***	0.57 (0.04,1.10)	0.30 (0.08,0.51)	1.42 (0.99,1.86)	48.2% (34.6%-59.0%)	816 (586-999)
P2T1	1.10 (0.89,1.36)	1.44 (1.29,1.62)***	1.84 (1.46,2.32)***	0.30 (-0.22,0.81)	0.16 (-0.09,0.41)	1.19 (0.84,1.55)	52.6% (37.9%-63.8%)	614 (443-745)
P2T2	0.99 (0.80,1.22)	1.26 (1.09,1.45)**	1.77 (1.39,2.26)***	0.53 (0.02,1.03)	0.30 (0.07,0.53)	1.42 (0.96,1.89)	45.7% (31.5%-56.9%)	862 (595-1075)
P2T3	1.09 (0.88,1.35)	1.65 (1.39,1.95)***	1.99 (1.51,2.60)***	0.25 (-0.40,0.90)	0.12 (-0.18,0.42)	1.14 (0.75,1.53)	43.5% (28.0%-55.7%)	651 (418-833)
P3T1	1.29 (1.02,1.63)*	1.43 (1.28,1.60)***	2.14 (1.63,2.80)***	0.42 (-0.25,1.09)	0.20 (-0.07,0.47)	1.24 (0.83,1.66)	49.7% (34.1%-61.7%)	524 (359-650)
P3T2	1.14 (0.90,1.44)	1.25 (1.09,1.42)**	2.15 (1.62,2.85)***	0.76 (0.08,1.45)	0.36 (0.12,0.59)	1.55 (0.99,2.11)	53.3% (38.8%-64.3%)	688 (501-831)
P3T3	1.23 (0.98,1.55)	1.59 (1.35,1.87)***	2.30 (1.62,3.26)***	0.49 (-0.41,1.38)	0.21 (-0.11,0.53)	1.27 (0.75,1.79)	53.5% (38.3%-64.9%)	586 (420-712)
P4T1	1.57 (1.04,2.37)*	1.42 (1.27,1.58)***	2.85 (1.95,4.17)***	0.86 (-0.41,2.14)	0.30 (-0.05,0.66)	1.43 (0.71,2.16)	56.5% (38.3%-69.4%)	426 (289-523)
P4T2	1.46 (0.99,2.16)	1.25 (1.10,1.43)***	2.45 (1.65,3.63)***	0.73 (-0.40,1.87)	0.30 (-0.07,0.67)	1.43 (0.68,2.18)	64.9% (48.7%-76.0%)	526 (394-616)
P4T3	1.70 (1.15,2.51)**	1.61 (1.37,1.90)***	2.19 (1.29,3.72)**	-0.13 (-1.49,1.23)	-0.06 (-0.71,0.59)	0.94 (0.36,1.52)	59.2% (39.5%-72.5%)	433 (289-531)

Note: RR¹: cumulative relative risk for PM_{2.5}-only events; RR²: cumulative relative risk for cold waves-only events; RR³ :
cumulative relative risk for joint exposure events to PM_{2.5} and cold waves. **P* < 0.05, ***P* < 0.01, ****P* < 0.001.

Table.S14 Cumulative Relative Risks and Interaction Indicators of PM_{2.5}-only, Cold-only, and Joint Exposures for Low,edu

Exposure Combination Definition	RR ¹ (95%CI)	RR ² (95%CI)	RR ³ (95%CI)	RERI (95%CI)	AP (95%CI)	S (95%CI)	AF (95%CI)	AN (95%CI)
P1T1	1.11 (0.93,1.32)	1.37 (1.22,1.54)***	1.94 (1.58,2.38)***	0.46 (-0.01,0.93)	0.24 (0.03,0.44)	1.31 (0.96,1.66)	48.5% (36.7%-58.0%)	1487 (1127-1780)
P1T2	1.08 (0.90,1.28)	1.28 (1.11,1.47)***	2.12 (1.69,2.65)***	0.76 (0.22,1.30)	0.36 (0.17,0.55)	1.56 (1.10,2.02)	52.8% (40.9%-62.3%)	1229 (953-1450)
P1T3	1.19 (1.00,1.43)*	1.71 (1.45,2.03)***	2.27 (1.77,2.92)***	0.37 (-0.31,1.04)	0.16 (-0.10,0.42)	1.19 (0.82,1.57)	55.9% (43.4%-65.7%)	896 (696-1053)
P2T1	1.07 (0.87,1.31)	1.40 (1.25,1.56)***	1.98 (1.59,2.47)***	0.52 (0.00,1.04)	0.26 (0.05,0.48)	1.36 (0.96,1.75)	49.5% (37.1%-59.5%)	1224 (916-1471)
P2T2	0.97 (0.79,1.19)	1.30 (1.13,1.49)***	1.93 (1.54,2.43)***	0.67 (0.15,1.19)	0.35 (0.14,0.55)	1.53 (1.05,2.01)	48.2% (34.9%-58.8%)	979 (709-1193)
P2T3	1.11 (0.90,1.37)	1.73 (1.47,2.04)***	2.11 (1.63,2.72)***	0.26 (-0.39,0.91)	0.12 (-0.16,0.41)	1.14 (0.77,1.51)	52.6% (38.8%-63.3%)	749 (552-901)
P3T1	1.20 (0.95,1.51)	1.39 (1.24,1.55)***	2.25 (1.74,2.91)***	0.67 (0.01,1.33)	0.30 (0.07,0.52)	1.42 (0.96,1.88)	55.6% (42.5%-65.6%)	955 (731-1128)
P3T2	1.06 (0.84,1.33)	1.29 (1.13,1.47)***	2.27 (1.74,2.96)***	0.93 (0.25,1.60)	0.41 (0.20,0.61)	1.69 (1.11,2.27)	55.9% (42.5%-66.2%)	835 (635-988)
P3T3	1.22 (0.97,1.52)	1.69 (1.43,1.98)***	2.27 (1.64,3.15)***	0.37 (-0.47,1.20)	0.16 (-0.16,0.48)	1.19 (0.73,1.65)	55.9% (38.9%-68.2%)	566 (394-690)
P4T1	1.60 (1.06,2.41)*	1.40 (1.25,1.56)***	3.08 (2.15,4.42)***	1.09 (-0.21,2.38)	0.35 (0.03,0.67)	1.54 (0.78,2.30)	67.5% (53.4%-77.4%)	731 (578-837)
P4T2	1.45 (0.98,2.15)	1.32 (1.16,1.51)***	2.56 (1.76,3.71)***	0.78 (-0.34,1.91)	0.31 (-0.04,0.65)	1.44 (0.72,2.17)	60.9% (43.3%-73.1%)	608 (432-729)
P4T3	1.96 (1.34,2.88)***	1.76 (1.49,2.06)***	2.01 (1.22,3.29)**	-0.71 (-1.99,0.57)	-0.35 (-1.14,0.43)	0.74 (0.31,1.16)	50.2% (18.3%-69.7%)	313 (114-434)

Note: RR¹: cumulative relative risk for PM_{2.5}-only events; RR²: cumulative relative risk for cold waves-only events; RR³ : cumulative relative risk for joint exposure events to PM_{2.5} and cold waves. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

Table.S15 Cumulative Relative Risks, Attributable Burden Metrics and Interaction Indicators of
PM_{2.5}-only, Cold-only, and Joint Exposures for High,edu

Exposure Combination Definition	RR ¹ (95%CI)	RR ² (95%CI)	RR ³ (95%CI)	RERI (95%CI)	AP (95%CI)	S (95%CI)	AF (95%CI)	AN (95%CI)
P1T1	1.02 (0.78,1.32)	1.61 (1.34,1.93)***	1.05 (0.77,1.43)	-0.57 (-1.08,-0.06)	-0.54 (-1.15,0.06)	0.65 (0.39,0.90)	4.8% (-29.8%-30.1%)	45 (-283-286)
P1T2	0.95 (0.73,1.23)	1.35 (1.11,1.63)**	1.02 (0.73,1.45)	-0.27 (-0.77,0.23)	-0.26 (-0.82,0.29)	0.79 (0.44,1.14)	2.0% (-38.2%-30.4%)	13 (-262-209)
P1T3	1.03 (0.80,1.33)	1.61 (1.28,2.02)***	1.05 (0.70,1.58)	-0.59 (-1.21,0.04)	-0.56 (-1.33,0.21)	0.64 (0.33,0.96)	4.8% (-43.1%-36.6%)	23 (-209-178)
P2T1	0.81 (0.60,1.08)	1.52 (1.28,1.81)***	1.00 (0.72,1.39)	-0.32 (-0.81,0.16)	-0.32 (-0.88,0.24)	0.76 (0.44,1.08)	0.0% (-38.9%-28.0%)	0 (-317-228)
P2T2	0.72 (0.54,0.96)*	1.28 (1.06,1.55)*	0.96 (0.67,1.36)	-0.05 (-0.51,0.42)	-0.05 (-0.55,0.45)	0.95 (0.50,1.41)	-4.2% (-48.4%-26.9%)	-26 (-304-169)
P2T3	0.81 (0.61,1.08)	1.51 (1.21,1.89)***	1.04 (0.69,1.57)	-0.28 (-0.88,0.31)	-0.27 (-0.93,0.39)	0.79 (0.38,1.19)	3.8% (-45.0%-36.3%)	18 (-210-169)
P3T1	0.77 (0.56,1.07)	1.52 (1.29,1.80)***	0.92 (0.62,1.36)	-0.38 (-0.88,0.13)	-0.41 (-1.09,0.26)	0.71 (0.37,1.05)	-8.7% (-61.0%-26.6%)	-46 (-322-140)
P3T2	0.74 (0.54,1.01)	1.31 (1.09,1.57)**	0.92 (0.61,1.41)	-0.12 (-0.63,0.39)	-0.13 (-0.72,0.46)	0.88 (0.42,1.35)	-8.7% (-65.3%-28.5%)	-39 (-290-127)
P3T3	0.82 (0.60,1.11)	1.56 (1.25,1.94)***	0.90 (0.52,1.56)	-0.47 (-1.12,0.18)	-0.52 (-1.48,0.43)	0.66 (0.24,1.07)	-11.1% (-92.5%-35.8%)	-37 (-305-118)
P4T1	0.59 (0.35,1.00)	1.49 (1.27,1.76)***	0.90 (0.51,1.61)	-0.18 (-0.84,0.47)	-0.20 (-1.02,0.62)	0.83 (0.26,1.40)	-11.1% (-97.4%-37.5%)	-41 (-360-139)
P4T2	0.58 (0.35,0.94)*	1.31 (1.09,1.56)**	0.82 (0.45,1.51)	-0.06 (-0.68,0.55)	-0.08 (-0.86,0.71)	0.93 (0.25,1.61)	-22.0% (-123.4%-33.4%)	-74 (-413-112)
P4T3	0.68 (0.41,1.12)	1.56 (1.25,1.93)***	0.78 (0.34,1.80)	-0.46 (-1.26,0.35)	-0.59 (-2.05,0.88)	0.63 (0.05,1.21)	-28.2% (-195.0%-44.3%)	-67 (-462-105)

Note: RR¹: cumulative relative risk for PM_{2.5}-only events; RR²: cumulative relative risk for cold waves-only events; RR³ : cumulative relative risk for joint exposure events to PM_{2.5} and cold waves. **P* < 0.05, ***P* < 0.01, ****P* < 0.001.

Table.S16 Cumulative Relative Risks, Attributable Burden Metrics and Interaction Indicators of
PM_{2.5}-only, Cold-only, and Joint Exposures for Male

Exposure Combination Definition	RR ¹ (95%CI)	RR ² (95%CI)	RR ³ (95%CI)	RERI (95%CI)	AP (95%CI)	S (95%CI)	AF (95%CI)	AN (95%CI)
P1T1	1.16 (0.96,1.40)	1.37 (1.21,1.56)***	1.83 (1.47,2.28)***	0.31 (-0.19,0.80)	0.17 (-0.07,0.41)	1.20 (0.86,1.54)	45.4% (31.9%-56.1%)	1069 (753-1323)
P1T2	1.12 (0.92,1.35)	1.27 (1.09,1.47)**	2.01 (1.58,2.56)***	0.63 (0.07,1.19)	0.31 (0.10,0.53)	1.45 (1.00,1.91)	50.2% (36.7%-60.9%)	902 (658-1093)
P1T3	1.23 (1.02,1.48)*	1.66 (1.39,1.99)***	2.20 (1.68,2.89)***	0.31 (-0.40,1.02)	0.14 (-0.15,0.43)	1.16 (0.77,1.56)	54.5% (40.4%-65.3%)	676 (500-810)
P2T1	1.07 (0.86,1.34)	1.39 (1.23,1.57)***	1.87 (1.47,2.37)***	0.41 (-0.12,0.94)	0.22 (-0.02,0.46)	1.28 (0.88,1.68)	46.5% (32.1%-57.9%)	897 (619-1117)
P2T2	0.97 (0.78,1.21)	1.27 (1.10,1.47)**	1.86 (1.45,2.38)***	0.62 (0.08,1.16)	0.33 (0.11,0.56)	1.50 (0.99,2.00)	46.2% (31.1%-58.0%)	732 (493-919)
P2T3	1.11 (0.89,1.38)	1.67 (1.41,2.00)***	2.05 (1.56,2.71)***	0.27 (-0.41,0.95)	0.13 (-0.17,0.44)	1.15 (0.75,1.55)	51.2% (35.7%-63.0%)	573 (399-704)
P3T1	1.20 (0.94,1.54)	1.38 (1.23,1.55)***	2.02 (1.53,2.66)***	0.44 (-0.22,1.09)	0.22 (-0.06,0.49)	1.28 (0.83,1.72)	50.5% (34.7%-62.5%)	678 (466-839)
P3T2	1.06 (0.84,1.35)	1.27 (1.11,1.47)***	2.09 (1.57,2.78)***	0.75 (0.08,1.43)	0.36 (0.12,0.60)	1.56 (0.99,2.14)	52.2% (36.3%-64.0%)	603 (420-741)
P3T3	1.20 (0.95,1.52)	1.64 (1.38,1.95)***	2.18 (1.53,3.11)***	0.34 (-0.53,1.22)	0.16 (-0.19,0.51)	1.19 (0.69,1.68)	54.1% (34.6%-67.8%)	437 (280-548)
P4T1	1.58 (1.03,2.42)*	1.39 (1.23,1.56)***	2.69 (1.83,3.97)***	0.73 (-0.53,1.98)	0.27 (-0.11,0.65)	1.37 (0.65,2.09)	62.8% (45.2%-74.8%)	548 (395-652)
P4T2	1.38 (0.92,2.08)	1.30 (1.13,1.49)***	2.28 (1.53,3.40)***	0.60 (-0.49,1.68)	0.26 (-0.13,0.65)	1.35 (0.63,2.07)	56.1% (34.6%-70.6%)	454 (280-571)
P4T3	1.82 (1.22,2.72)**	1.71 (1.44,2.03)***	1.91 (1.11,3.28)*	-0.62 (-1.92,0.68)	-0.33 (-1.15,0.50)	0.75 (0.28,1.23)	47.6% (10.0%-69.5%)	251 (53-366)

Note: RR¹: cumulative relative risk for PM_{2.5}-only events; RR²: cumulative relative risk for cold waves-only events; RR³ : cumulative relative risk for joint exposure events to PM_{2.5} and cold waves. **P* < 0.05, ***P* < 0.01, ****P* < 0.001.

Table.S17 Cumulative Relative Risks, Attributable Burden Metrics and Interaction Indicators of
PM_{2.5}-only, Cold-only, and Joint Exposures for Female

Exposure Combination Definition	RR ¹ (95%CI)	RR ² (95%CI)	RR ³ (95%CI)	RERI (95%CI)	AP (95%CI)	S (95%CI)	AF (95%CI)	AN (95%CI)
P1T1	1.00 (0.82,1.21)	1.48 (1.29,1.69)***	1.60 (1.27,2.01)***	0.12 (-0.34,0.58)	0.08 (-0.20,0.35)	1.08 (0.76,1.40)	37.5% (21.4%-50.3%)	623 (355-836)
P1T2	0.97 (0.80,1.18)	1.34 (1.15,1.56)***	1.69 (1.31,2.18)***	0.38 (-0.13,0.89)	0.23 (-0.03,0.48)	1.29 (0.86,1.72)	40.8% (23.7%-54.1%)	498 (289-660)
P1T3	1.08 (0.89,1.31)	1.76 (1.46,2.12)***	1.77 (1.32,2.36)***	-0.07 (-0.72,0.57)	-0.04 (-0.41,0.33)	0.96 (0.62,1.30)	43.5% (24.5%-57.7%)	369 (208-490)
P2T1	0.92 (0.73,1.15)	1.47 (1.29,1.67)***	1.60 (1.25,2.05)***	0.21 (-0.27,0.70)	0.13 (-0.14,0.41)	1.15 (0.78,1.52)	37.5% (20.0%-51.2%)	509 (271-695)
P2T2	0.84 (0.67,1.06)	1.33 (1.15,1.56)***	1.54 (1.19,2.00)**	0.36 (-0.13,0.85)	0.24 (-0.03,0.51)	1.31 (0.85,1.77)	35.1% (15.8%-49.9%)	377 (170-537)
P2T3	0.96 (0.77,1.21)	1.74 (1.45,2.08)***	1.65 (1.23,2.21)***	-0.05 (-0.67,0.57)	-0.03 (-0.41,0.35)	0.97 (0.61,1.33)	39.4% (18.8%-54.8%)	304 (145-423)
P3T1	0.97 (0.75,1.25)	1.45 (1.29,1.64)***	1.81 (1.36,2.42)***	0.39 (-0.21,1.00)	0.22 (-0.07,0.50)	1.28 (0.82,1.74)	44.8% (26.3%-58.6%)	405 (238-530)
P3T2	0.88 (0.68,1.13)	1.33 (1.15,1.54)***	1.80 (1.33,2.44)***	0.59 (-0.03,1.21)	0.33 (0.07,0.59)	1.48 (0.91,2.06)	44.4% (24.8%-59.0%)	347 (193-460)
P3T3	1.00 (0.78,1.27)	1.72 (1.44,2.05)***	1.71 (1.17,2.50)**	-0.00 (-0.76,0.75)	-0.00 (-0.44,0.44)	1.00 (0.56,1.44)	41.5% (14.5%-60.0%)	222 (78-320)
P4T1	1.03 (0.66,1.59)	1.46 (1.29,1.65)***	2.34 (1.55,3.53)***	0.85 (-0.23,1.93)	0.36 (0.03,0.70)	1.57 (0.75,2.40)	57.3% (35.5%-71.7%)	332 (206-416)
P4T2	1.00 (0.66,1.53)	1.36 (1.18,1.57)***	1.96 (1.28,3.00)**	0.59 (-0.36,1.55)	0.30 (-0.08-0.68)	1.44 (0.65,2.22)	49.0% (21.9%-66.7%)	256 (114-349)
P4T3	1.27 (0.83,1.92)	1.76 (1.48,2.10)***	1.53 (0.86,2.71)	-0.50 (-1.57,0.57)	-0.33 (-1.19,0.53)	0.75 (0.27,1.24)	34.6% (-16.0%-63.2%)	115 (-53-210)

Note: RR¹: cumulative relative risk for PM_{2.5}-only events; RR²: cumulative relative risk for cold waves-only events; RR³ : cumulative relative risk for joint exposure events to PM_{2.5} and cold waves. **P* < 0.05, ***P* < 0.01, ****P* < 0.001.

Table.S18 Sensitivity analysis comparing cumulative relative risks (CRR, lag 0–21) for joint exposure under the main analysis (combination-specific reference) and a uniform reference category (PM_{2.5} < 115 µg/m³ and temperature > -0.4°C), by cause of death

Exposure	Non-accidental Death		Circulatory disease Mortality		Respiratory disease Mortality	
Combination	Main analysis	Uniform reference	Main analysis (CRR,	Uniform reference	Main analysis (CRR,	Uniform reference
Definition	(CRR, 95% CI)	(CRR, 95% CI)	95% CI)	(CRR, 95% CI)	95% CI)	(CRR, 95% CI)
P1T1	1.73 (1.43-2.10)	1.93 (1.57-2.37)	1.81 (1.44,2.26)	1.97 (1.54-2.50)	1.36(0.93,1.98)	1.55 (1.03-2.34)
P1T2	1.87 (1.52-2.31)	1.87 (1.51-2.32)	1.99 (1.56,2.55)	1.97 (1.52-2.54)	1.53 (1.00,2.35)	1.66 (1.08-2.57)
P1T3	2.01 (1.59-2.55)	1.98 (1.55-2.52)	2.24 (1.70,2.97)	2.08 (1.55-2.78)	1.48(0.91,2.41)	1.83 (1.12-3.00)
P2T1	1.75 (1.43-2.15)	2.53 (1.76-3.63)	1.86 (1.46,2.37)	2.55 (1.66-3.91)	1.48(0.98,2.25)	2.34 (1.13-4.84)

P2T2	1.72 (1.39-2.13)	2.00 (1.60-2.49)	1.84 (1.42,2.37)	2.07 (1.60-2.68)	1.49(0.96,2.31)	1.73 (1.10-2.70)
P2T3	1.88 (1.48-2.39)	1.84 (1.47-2.30)	2.10 (1.58,2.79)	1.94 (1.49-2.52)	1.42(0.86,2.33)	1.70 (1.08-2.69)
P3T1	1.94 (1.53-2.46)	2.01 (1.55-2.59)	2.03 (1.52,2.69)	2.08 (1.53-2.81)	1.76 (1.08,2.86)	1.99 (1.19-3.35)
P3T2	1.97 (1.54-2.53)	2.22 (1.52-3.23)	2.03 (1.51,2.73)	2.12 (1.36-3.30)	1.91 (1.15,3.18)	2.45 (1.15-5.19)
P3T3	1.98 (1.46-2.70)	2.09 (1.62-2.70)	2.15 (1.49,3.10)	2.29 (1.70-3.09)	1.57(0.83,2.98)	1.61 (0.95-2.73)
P4T1	2.55 (1.82-3.57)	1.93 (1.48-2.50)	2.62 (1.76,3.92)	2.14 (1.58-2.91)	2.34 (1.18,4.65)	1.56 (0.92-2.65)
P4T2	2.15 (1.52-3.05)	2.07 (1.48-2.89)	2.12 (1.40,3.21)	2.21 (1.49-3.27)	2.33 (1.15,4.73)	1.59 (0.81-3.12)
P4T3	1.75 (1.10-2.80)	2.05 (1.19-3.51)	1.83(1.05,3.20)	1.97 (1.05-3.72)	1.48(0.56,3.87)	1.53 (0.51-4.52)

Note:CRR = cumulative relative risk over lag 0–21 days. Main analysis used combination-specific reference days (no PM_{2.5} event and no cold wave event according to the respective thresholds). Uniform reference used fixed criteria: PM_{2.5} < 115 µg/m³ and temperature > -0.4°C.

Table.S19 Sensitivity analysis excluding the COVID-19 policy transition period and winsorizing daily deaths: cumulative relative risks (CRR, 95% CI) for joint exposure

Exposure Combination Definition	Non-accidental Death			Circulatory disease Mortality			Respiratory disease Mortality		
	Main	Excluding COVID	Winsorized	Main	Excluding COVID	Winsorized	Main	Excluding COVID	Winsorized
P1T1	1.73 (1.43-2.10)	1.36 (1.16-1.60)	1.67 (1.42-1.97)	1.81 (1.44-2.26)	1.38 (1.13-1.69)	1.69 (1.39-2.06)	1.36 (0.93-1.98)	1.10 (0.74-1.63)	1.44 (1.01-2.06)
P1T2	1.87 (1.52-2.31)	1.47 (1.23-1.74)	1.81 (1.51-2.16)	1.99 (1.56-2.55)	1.52 (1.23-1.90)	1.88 (1.52-2.34)	1.53 (1.00-2.35)	1.22 (0.79-1.89)	1.63 (1.09-2.44)
P1T3	2.01 (1.59-2.55)	1.51 (1.25-1.83)	1.90 (1.55-2.33)	2.24 (1.70-2.97)	1.64 (1.28-2.09)	2.07 (1.62-2.64)	1.48 (0.91-2.41)	1.14 (0.70-1.86)	1.57 (0.99-2.50)
P2T1	1.75 (1.43-2.15)	1.32 (1.12-1.57)	1.66 (1.39-1.98)	1.86 (1.46-2.37)	1.35 (1.09-1.67)	1.74 (1.40-2.16)	1.48 (0.98-2.25)	1.26 (0.83-1.93)	1.52 (1.02-2.26)
P2T2	1.72 (1.39-2.13)	1.42 (1.19-1.69)	1.68 (1.40-2.02)	1.84 (1.42-2.37)	1.48 (1.18-1.85)	1.78 (1.43-2.23)	1.49 (0.96-2.31)	1.33 (0.86-2.07)	1.62 (1.07-2.44)
P2T3	1.88 (1.48-2.39)	1.44 (1.18-1.75)	1.77 (1.44-2.17)	2.10 (1.58-2.79)	1.58 (1.23-2.02)	1.96 (1.53-2.52)	1.42 (0.86-2.33)	1.17 (0.71-1.91)	1.52 (0.95-2.44)
P3T1	1.94 (1.53-2.46)	1.45 (1.19-1.76)	1.76 (1.43-2.17)	2.03 (1.52-2.69)	1.48 (1.15-1.90)	1.86 (1.45-2.39)	1.76 (1.08-2.86)	1.42 (0.86-2.34)	1.77 (1.12-2.81)
P3T2	1.97 (1.54-2.53)	1.61 (1.32-1.98)	1.86 (1.50-2.30)	2.03 (1.51-2.73)	1.65 (1.27-2.14)	1.94 (1.50-2.52)	1.91 (1.15-3.18)	1.61 (0.96-2.70)	2.03 (1.25-3.27)
P3T3	1.98 (1.46-2.70)	1.66 (1.29-2.14)	1.91 (1.46-2.50)	2.15 (1.49-3.10)	1.73 (1.26-2.38)	2.06 (1.49-2.84)	1.57 (0.83-2.98)	1.23 (0.65-2.32)	1.71 (0.93-3.13)
P4T1	2.55 (1.82-3.57)	1.75 (1.32-2.31)	2.29 (1.71-3.06)	2.62 (1.76-3.92)	1.70 (1.19-2.43)	2.35 (1.65-3.35)	2.34 (1.18-4.65)	1.96 (0.97-3.95)	2.55 (1.33-4.87)
P4T2	2.15 (1.52-3.05)	1.72 (1.29-2.29)	2.08 (1.53-2.81)	2.12 (1.40-3.21)	1.61 (1.12-2.31)	2.04 (1.42-2.93)	2.33 (1.15-4.73)	1.95 (0.95-3.99)	2.60 (1.34-5.06)
P4T3	1.75 (1.10-2.80)	1.85 (1.26-2.72)	1.89 (1.26-2.84)	1.83 (1.05-3.20)	1.72 (1.05-2.80)	1.85 (1.13-3.03)	1.48 (0.56-3.87)	1.34 (0.51-3.51)	1.77 (0.71-4.39)

Note:CRR = cumulative relative risk over lag 0–21 days. Main analysis used the original dataset. Excluding COVID = analysis excluding December 2022 – February 2023 (the COVID-19 policy transition period). Winsorized = analysis with daily death counts winsorized at the 99th percentile. Values in parentheses are 95% confidence intervals.